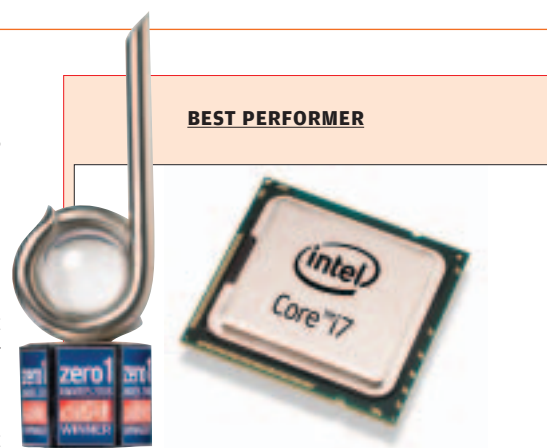


CPUs

Processors are the ramp models of the computing world. Nobody makes as much a splash entry as a new processor and although in our estimation much of the attention is overrated, one cannot deny the appeal of the latest and fastest dual- and quad-cores. Parallelism is the theme of the future for Intel and AMD as their CPUs have grown extra cores progressively. You needn't worry about your CPU behaving sluggishly anymore; the latest dual- and quad-cores are veritable powerhouses and can tackle the heaviest of applications with ease. The fact that most applications aren't multithreaded to begin with doesn't faze these manufacturers one bit; they've decided to approach the problem with brute force. Add an extra core, enlarge the cache and bump up the clock speeds, all while shrinking the die manufacturing process to cut costs.

Intel has shifted to a 45nm process while AMD is in the process of making the transition to 45nm. We've seen Intel move to DDR2 in the past and it seems history is about to repeat itself with the latest Intel CPU codenamed Nehalem. This time DDR3 is being adopted in lieu of DDR2. Although AMD hasn't planned such a move they may well have to consider integrating DDR3 support on their next generation Deneb CPUs; which are scheduled for an early 2009 release. AMD's own Phenom CPUs have done little to steal Intel's thunder in the performance arena and these remain budget options compared to Core 2 Duos and Core 2 Quads. Intel's quad-cores, also called Yorkfield CPUs, have served their purpose well and while they're still top performers it seems Nehalem is the CPU to watch. This CPU has an integrated memory controller which is a first for Intel; as they've totally relied on their FSB for memory to CPU communication. This has been identified as a bottleneck by us some years back, and it seems to be a logical move to an integrated memory controller. With the Nehalem, Intel shifts to an integrated memory controller, so the Northbridge no longer has memory communication to worry about. The Nehalem has grown in terms of pin count; from an LGA 775 to an LGA 1366-contact configuration. In addition Nehalem also has an L3 cache, although the L2 caches are much smaller than previous dual- and quad-cores. So now both AMD and Intel are on somewhat levelled playing fields in terms of technology.



BEST PERFORMER

Intel i7 965 Extreme (3.2 GHz)

A long name for such a mighty CPU and is not to be confused with the P965 motherboard chipset of yesteryear. Our performance winner in our test last month was initially the Intel Core 2 Extreme QX 9775 which ran at a mighty 3.2 GHz. This CPU supports an FSB of 1600 MHz and 12 MB of L2 cache. As impressive as that may seem, it pales in comparison to the newcomer who manages to outdo it in 90 per cent of all benchmarks. Bear in mind that the i7 965 Extreme was running slower memory with slower timings and the platform itself was in its initial stages; this is bound to get faster as support and drivers improve. In short Nehalem will get faster. This CPU also supports hyper-threading; which means Windows actually recognises it as an eight core processor — cool and immeasurable flaunt value. With an L2 cache of 1 MB, L3 cache of 8 MB and an HTT link that runs at a speed of 6.4 Gbps the i7 965 Extreme has a superb set of on-paper specifications

Whatever the application be it 3D modelling and rendering, video encoding, file compression/decompression, image editing or benchmarking the i7 965 Extreme is blazingly fast. Games aren't really optimised for multi-cores as yet but that should change in the future; yet we saw a definite upward scaling with shifting from a QX 9770 to the i7 965 Extreme, and barring a single bug while benchmarking *Unreal Tournament 3*, this is the CPU to get if you really want speed and aren't afraid to pay for it. The i7 965 Extreme should debut at a price of Rs 50,000 and will spawn a host of cheaper variants in the future.



REIGNING CHAMPION

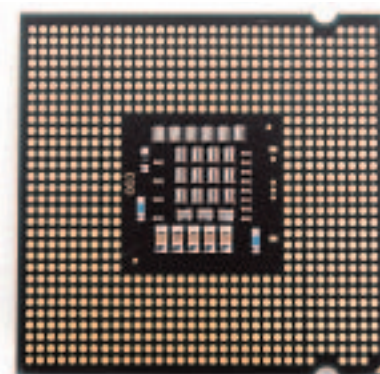
Intel Core 2 Q8200

Intel's quad-cores have been the object of desire by most performance junkies right from the time of their launch. Prices have fallen, better models have come out and in general buying a quad core is not an expensive proposition and one that even mid-range spenders can afford. All this started with the Q6600 (2.4 GHz), which was a bestseller for Intel. The Q8200 is an entry level quad core. At 2.33 GHz it features a meagre four megabytes of L2 cache (other quad cores have either eight or 12 MB of L2 cache). However, there's nothing entry-level about its performance and it lays the top-end AMD Phenom processors to rest rather easily. With a 1,333 MHz FSB this CPU is built around a cool running 45nm fabrication process and is very suitable for gamers and 3D professionals alike who want something powerful within price constraints. At Rs 10,950 or thereabouts the Q8200 is a superb buy for anyone looking for a powerful solution. So what is our only beef with Intel? The fact that they follow a global pricing structure and this CPU has got more expensive as the rupee falls further behind the dollar.

CHALLENGER(S)

Intel Core 2 Duo E8400

Intel had to try to beat themselves and the Core 2 Duo E8400 represents the best of that effort.



We'd tested the E8500 (3.16 GHz) earlier and at Rs 10,000 we thought that it was a little too expensive for a dual core, especially with the Q8200 and Q6600 lurking around. The E8400 is priced sensibly at Rs 8,500 and is a superb option for someone looking for a powerful PC or gamers. Given that games aren't multi-core optimised and respond to clock speed increments more, the E8400 is able to outpace some much costlier quad-cores that have slower clock speeds. This Wofdale CPU has 6 MB of L2 cache and runs at an FSB of 1333 MHz; the best part is that it's a 45nm part and runs very cool.

AND THE WINNER IS

Intel Core 2 Q8200



Despite the high clock speed on the Core 2 Duo E8400, the Core 2 Quad E8200 manages to steal its thunder. Having a PC with four cores is so much more future proof than a PC with two cores and the Core 2 Quad Q8200 is a solid choice for a powerful home PC. We reckon only gamers will really have any use for the higher clock speeds and opt for the E8400 over the Q8200. Even people working with 3D rendering will opt for the sheer parallel processing that four cores offer and the Q8200 is a superb choice.